

Changjian Zhou

Ph.D. Student · The University of Melbourne

Department of Infrastructure Engineering

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Research interests

Scientific machine learning and neural PDE solvers for sim-to-real transfer; computational geomechanics and constitutive model calibration; uncertainty quantification.

Education

Ph.D. · The University of Melbourne

2025-present · Department of Infrastructure Engineering

Supervisors: Negin Yousefpour and Guillermo A. Narsilio

M.Eng. in Civil and Hydraulic Engineering · Shanghai Jiao Tong University

2021-2024

Supervisor: Guanlin Ye

B.Eng. in Hydraulic and Hydropower Engineering

Hefei University of Technology · 2015-2019

Research experience

Research Assistant · Tsinghua University

2024-2025

Computational modelling of bone cement dispersion, with CT-based reconstruction and computational fluid dynamics.

Selected honours and awards

Yang-Yuqiu Scholarship, Shanghai Jiao Tong University · 2023

Second Prize, 19th China Postgraduate Mathematical Modelling Competition

Distinguished Undergraduate Thesis Award, Hefei University of Technology · 2019

Publications

Changjian Zhou

1. PhysGuard: Fisher-Guided Gradient Projection for Sim-to-Real Neural PDE Surrogates

Changjian Zhou, Junfeng Fang, Negin Yousefpour, Peng Wu, Bin Yan, Guillermo A. Narsilio

NeurIPS 2026 · Poster

arxiv.org/abs/2606.16602

2. A weak-form-based 1D numerical model for guided wave dispersion in subsea buried pipes embedded in saturated poroelastic soil

Anchen Ni, Wenbin Wei, Tao Zhuge, **Changjian Zhou**, Facheng Wang

Ocean Engineering, 363, 126622 (2026)

doi.org/10.1016/j.oceaneng.2026.126622

3. Stochastic Polynomial Surrogate Models for Uncertainty Quantification of Offshore Plate Anchor Capacity

Negin Yousefpour, Bo Wang, **Changjian Zhou**, Alessio Mentani

Geotechnical and Geological Engineering, 44, 305 (2026)

doi.org/10.1007/s10706-026-03824-0

4. A combined machine learning/search algorithm-based method for the identification of constitutive parameters from laboratory tests and in-situ tests

Changjian Zhou, Bin Gao, Bin Yan, Wenxuan Zhu, Guanlin Ye

Computers and Geotechnics, 170, 106268 (2024)

doi.org/10.1016/j.compgeo.2024.106268

Accepted manuscript

A KAN-GSA surrogate modelling framework to streamline the calibration of advanced constitutive models: A case study on the Shanghai Model

First author

Engineering Applications of Artificial Intelligence · Accepted